

Yilang Liu

✉ yilang.liu@yale.edu  [yilangliu.github.io](https://github.com/yilangliu)  [linkedin.com/in/yilangliu](https://www.linkedin.com/in/yilangliu)  Google Scholar

EDUCATION

Yale University

Ph.D. Candidate in Mechanical Engineering

M.S. & M.Phil in Mechanical Engineering (en route to Ph.D.)

New Haven, CT

Expected Graduation May 2027

Carnegie Mellon University

M.S. in Mechanical Engineering – Research Program

Pittsburgh, PA

May 2022

Chongqing University

B.E. in Mechanical Design, Manufacture and Automation

Chongqing, China

April 2020

University of Cincinnati

B.S. in Mechanical Engineering (Cum Laude)

Cincinnati, OH

April 2020

PUBLICATIONS

Haoxiang You, **Yilang Liu**, Ian Abraham. “[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)”. Advances in Neural Information Processing Systems (**Spotlight** NeurIPS 2025).

Yilang Liu, Haoxiang You, Ian Abraham. “[Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes](#)”. IEEE International Conference on Robotics and Automation (ICRA) 2026.

Christian Hughes, **Yilang Liu**, Julia Engdahl, Houston Warren, Darrick Lee, Fabio Ramos, Travis Miles, Ian Abraham. “[Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields](#)”. Robotics: Science and Systems (RSS) 2026.

Haoxiang You*, **Yilang Liu***, Ian Abraham. “[Efficient On-Policy Visual RL via Stochastic Decoupled Policy Gradient](#)”. CVPR 2026 Workshop on Bridging Vision, Language, and Action. (*Equal contribution.)

Yilang Liu, Haoxiang You. “[Hybrid Mode Sample-Based Control](#)”. RSS 2025 Workshop on Whole-Body Control and Bimanual Manipulation: Applications in Humanoids and Beyond, 2025.

Yilang Liu, Amir Barati Farimani. “[An Energy-Saving Snake Locomotion Pattern Learned in a Physically Constrained Environment With Online Model-Based Policy Gradient Method](#)”. ASME. J. Mechanisms Robotics, 2023.

Yilang Liu, Y. Chen, H. Li. “[Design an Augmentation Exoskeleton to Enhance Lifting Strength](#)”. ASME IMECE2020-24658, 2020.

RESEARCH EXPERIENCE

[Efficient On-Policy Visual RL via Stochastic Decoupled Policy Gradient](#)

New Haven, CT

Advisor: Prof. Ian Abraham

01/2026–Present

- Proposed **Stochastic Decoupled Policy Gradient (SDPG)**, a lightweight visual RL method that trains visuomotor control policies end-to-end in a **few hours on a single NVIDIA RTX 4080 GPU**.
- Estimated policy gradients via random perturbations of trajectory rollouts, eliminating full trajectory differentiation and requiring **orders of magnitude fewer** batch-rendered environments to reduce compute and memory overhead.
- Designed an adaptive exploration strategy and reward-invariant normalization for numerically stable updates, and developed a unified perspective connecting classical policy-gradient methods.
- Outperformed baselines (DrQv2, TD-MPC2, DreamerV3, PPO) on visual MuJoCo benchmarks in training time, memory usage, and reward.
- Released a suite of realistic egocentric visual RL benchmarks spanning dexterous manipulation and challenging locomotion, and demonstrated sim-to-real transfer on a Unitree Go2 robot.

[Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields](#)

New Haven, CT

Advisor: Prof. Ian Abraham

06/2025–01/2026

- Developed a flow-adaptive ergodic trajectory optimization framework for robotic exploration in time-varying environments using Maximum Mean Discrepancy (MMD) metrics.
- Derived a theoretical formulation of ergodic coverage on evolving domains, enabling coverage guarantees under dynamic flow fields.
- Designed and implemented a sampling-based trajectory optimization algorithm capable of planning exploration paths in non-convex and flow-constrained environments.
- Demonstrated the approach on multiple robotic platforms, including aerial drones (crazyflies) and Unitree Go2 quadruped robots, validating coverage performance in both simulation and real-world experiments.
- Achieved up to **42% improvement** in exploration performance over information-theoretic baselines in dynamic flow

environments.

Accelerating Visual-Policy Learning through Parallel Differentiable Simulation

New Haven, CT

Advisor: Prof. Ian Abraham

01/2025–06/2025

- Proposed **D.Va (Decoupled Visual-Based Analytical Policy Gradient)**, a novel method for accelerating visual policies by **10x wall-clock time** using parallel differentiable simulation.
- Proposed to decouple visual observations from computation graphs, eliminating differentiating through rendering and achieving training a humanoid running policy in **4 hours** on a single GPU.
- Implemented visual policy learning method from differentiable rendering as baseline comparison using PyTorch3D and proving smoother first-order policy gradient without differentiating through render.
- Benchmarked model-free and model-based visual policy learning approaches and showed **5x higher rewards** in humanoid running tasks against existing algorithms.
- Provided an empirical analysis of analytical policy gradients and highlighted the new opportunities for integrating policy learning with trajectory optimization techniques.

Sample-Based Hybrid Mode Control

New Haven, CT

Advisor: Prof. Ian Abraham

12/2024–04/2025

- Designed a sample-based control strategy for high-dimensional long-horizon tasks using hybrid control theory, achieving complex mode switching behaviors.
- Analyzed theoretical correctness and proved performance guarantees on optimizing mode sequencing.
- Achieved **85% higher reward** in manipulation and **41% in humanoid locomotion** vs. baselines, enabling contact-rich in-hand manipulation.
- Implemented state estimation module for real-world Unitree Go2 deployment based on an extended Kalman filter using proprioception information.
- Deployed **real-time torque-level** agile locomotion of a Unitree Go2 quadruped capable of switching from footstand to handstand within sample-based predictive control framework.

PROFESSIONAL EXPERIENCE

Dexmate Inc.

Santa Clara, CA

Robotics Intern

10/2025–01/2026

- Deployed the Vision-Language-Action (VLA) model **GR00T N1.6** on **Dexmate Vega**, a dual-arm mobile manipulator, for an Amazon-designated project, executing a long-horizon (**2-minute**) bimanual task—grasping a PVC box by its handle with one gripper while expanding it with the other, then placing it aside—achieving a **100% success rate**.
- Implemented the VLA model $\pi_{0.5}$ for a client project, deploying it on the physical robot for a bimanual task placing a condenser onto a conveyor; smoothed action-chunk transitions with Real-Time Action Chunking (RTC) in PyTorch for stable execution at ~20 Hz, achieving a **100% success rate** validated in factory testing.
- Deployed the Action Chunking with Transformers (ACT) imitation-learning algorithm on the physical robot using a bimanual teleoperated dataset, achieving reliable bottle grasping and placing from vision.
- Trained a reinforcement learning policy with Proximal Policy Optimization (PPO) in simulation (Genesis and Isaac Sim), guided by RRT motion planning, achieving a **100% success rate** on cube grasping with a 6-DoF dexterous hand.

TEACHING

Teaching Fellow – ENAS 773: Introduction to Robotics, Control, and Learning

Yale University

- Led weekly recitations and office hours, advised student final projects, and assessed coursework on robotics, control, and learning.

Teaching Assistant – 18-461: Introduction to Machine Learning for Engineers

Carnegie Mellon University

- Guided students through machine learning fundamentals via recitations and office hours, mentored course projects, and graded assignments and exams.

ACADEMIC SERVICE

Conference Reviewer: ICRA (2022–2026), IROS (2022–2026), RSS (2025), ECCV (2024).

Journal Reviewer: IEEE Robotics and Automation Letters (RA-L).

SKILLS

Programming & ML: Python, C++, PyTorch, PyTorch3D.

Robotics & Tools: MuJoCo, Isaac Gym, Isaac Sim, Genesis, Brax; ROS 1/2; Unitree A1, Unitree Go2; LeRobot, Weights & Biases.